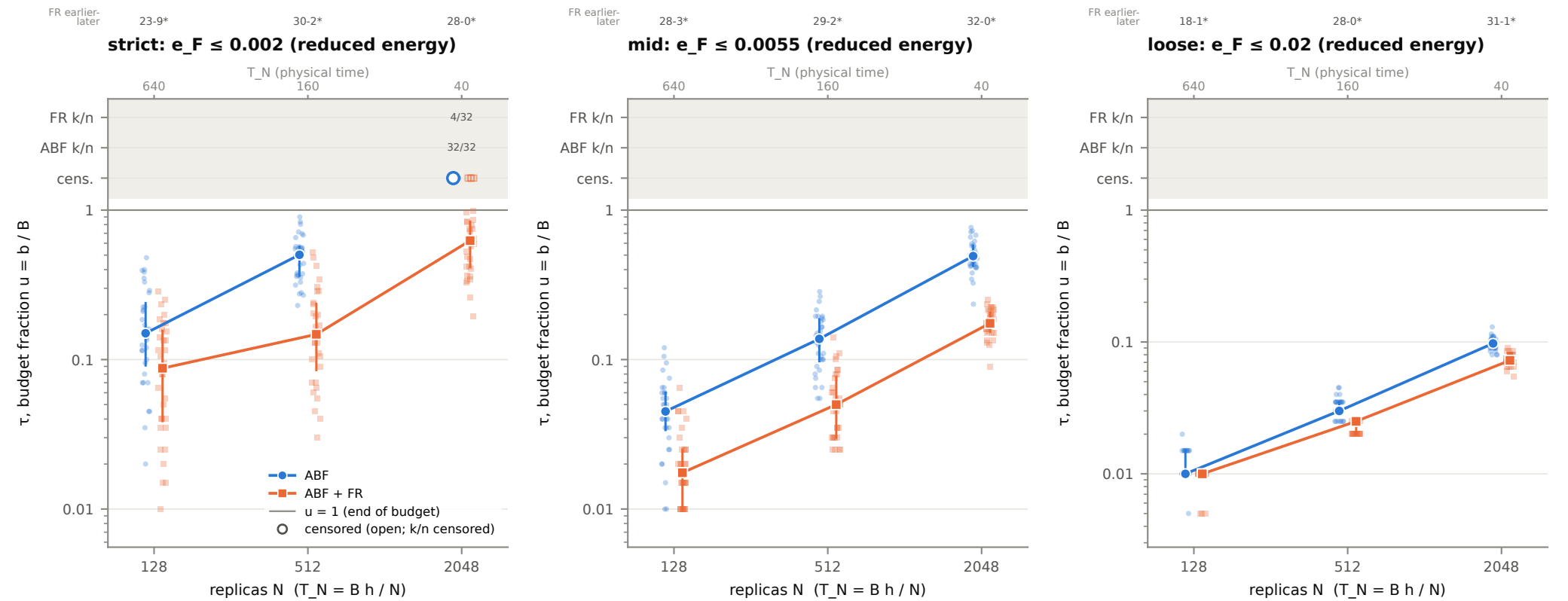


gateway alpha0.5: persistent time to free-energy accuracy, budget fraction

Equal total budget per arm per seed: $B = N \times n_steps = 3.2768e+09$ walker-steps at every N ($h = 2.5e-05$, $T_N = B \cdot h / N$); 32 seeds per (N , method). $\tau(\epsilon)$ = first save after which $e_F \leq \epsilon$ at every later save; never met: censored ($> T_N$, > 1).



gateway alpha0.5 · S4b_tau_free_energy_budget · status: all planned (N , seed, method) complete · FR is not defined at $N = 1$ (ABF only, by design).
Markers: seed median (large open marker in the 'cens.' band = median censored); bars: IQR; faint: seeds; censored seeds are open markers in the band, never success at T_N .
Row above each panel: the PAIRED comparison at that N , seeds with FR earlier - FR later than ABF (finite beats censored, both censored = tie; * = exact sign test $p < 0.05$; medians, G and CIs in summary.json / tables.md). Budget axis: tau on the common budget grid (200 uniform + 24 log-spaced u , identical at every N) so that N are compared at equal resolution.
Source results/mechanism/matched_free_energy/alpha0.5/analysis/summary.json (generated 2026-10-10T14:19:19Z, eqb_metrics/2, 10000 bootstrap resamples); plot_synthesis/2.